

Rethinking What to Communicate for Collaborative Perception: A Parameter-Centric Perspective

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Abstract—Collaborative perception enhances autonomous driving by allowing agents to share complementary perceptual information. Existing methods typically adopt a Feature-Centric Communication (FCC) perspective that focuses on selecting or compressing feature representations for transmission. A shared representational basis offers an opportunity to reduce online communication by encoding observations through compact parameters. Building on this idea, we introduce a Parameter-Centric Communication (PCC) perspective, in which these parameters specify the composition of representational bases retained locally at each agent. We instantiate PCC as ParaComm, a learnable dictionary-driven framework for communication-efficient collaborative perception. In ParaComm, continuous parameters learned from BEV features specify how shared dictionary atoms are combined to form the current collaborative representation. To reduce transmission volume, ParaComm progressively sparsifies these parameters across spatial locations and dictionary atoms and further compacts them through parameter sharing and low-bit quantization. The receiver then explicitly reconstructs collaborative features using the transmitted parameters and its local copy of the shared dictionary. Extensive experiments on OPV2V, V2XSet, and DAIR-V2X validate the effectiveness and generality of ParaComm. On OPV2V, for example, it reduces the communication volume from 34.375 MB to 99.688 KB, corresponding to over $350\times$ compression, while still improving detection accuracy relative to standard full-feature collaboration.

Index Terms—Collaborative Perception, Communication-Efficient Collaboration, Autonomous Driving.

I. INTRODUCTION

ADVANCES in mobile computing and edge infrastructure are fostering collaboration among distributed agents for increasingly complex perception tasks [1]. In autonomous driving, vehicles and roadside units (RSUs) exchange and fuse observations beyond their individual sensing coverage through collaborative perception [2]–[4]. The resulting multi-view information better reveals distant and occluded targets [5]–[7] by

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Project page: <https://weiquanmin.github.io/ParaComm/>.

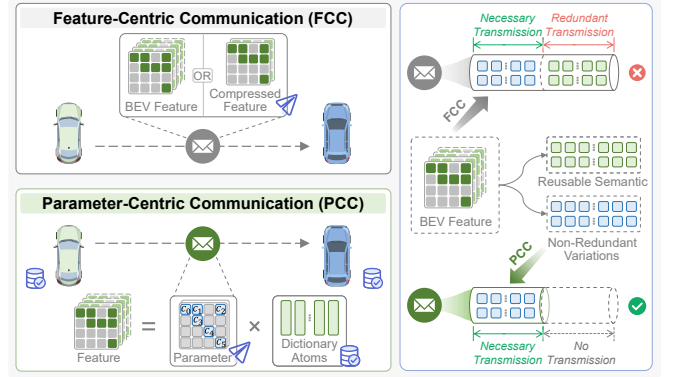


Fig. 1. Feature-Centric Communication (FCC) versus Parameter-Centric Communication (PCC). FCC transmits features or their compressed feature representations, which still mix shared semantic bases with observation-specific variations. PCC instead keeps the shared bases locally through dictionary atoms and communicates only sparse parameters for the latter. Therefore, PCC concentrates communication resources on the most informative differences for collaboration.

alleviating restricted fields of view and partial observability [8]–[10]. These capabilities are critical for safe autonomous driving in complex scenarios such as intersections and dense urban traffic [11]. However, the trade-off between perceptual performance and communication bandwidth remains a major bottleneck for practical collaborative perception because of the limitations of wireless links.

To achieve efficiency, communication-efficient methods primarily rely on selective transmission or compact encoding of intermediate feature representations [12]. For example, Where2comm communicates informative spatial regions [13], while InfoCom retains feature content according to task relevance [14]. Despite these implementation differences, they share the same fundamental design principle: intermediate features or their compressed variants are packed as a whole into online messages. We refer to this prevailing communication perspective as **Feature-Centric Communication (FCC)**. Such packaging typically conflates two qualitatively distinct components. One component reflects a reusable semantic base, such as relatively stable geometric and semantic patterns associated with common object categories. The other captures observation-specific variation induced by the current viewpoint and sensing conditions, such as visibility, occlusion, and local object configuration. Under FCC, part of the communication budget is still spent on the repeated transmission of reusable information that could otherwise remain local.

This inherent redundancy opens up a new opportunity to

reduce communication overhead. The key observation is that online communication and model deployment unfold on different time scales: online messages are exchanged continuously during collaboration, whereas the collaborative model remains resident locally for long-term reuse. Therefore, reusable semantic information can be organized into a shared representational basis and preloaded onto each agent at deployment. Each online message then only needs to convey compact parameters that are derived from the current BEV features and characterize the corresponding observation with respect to this basis. This yields a **Parameter-Centric Communication (PCC)** perspective of collaborative communication. A comparison between FCC and PCC is illustrated in Fig. 1. Codebook-based methods move in a related direction by maintaining pre-shared feature codewords across agents and transmitting only compact codeword indices [15], [16]. However, the online message in these methods is used to retrieve predefined feature proxies, so the representation available at the receiver remains limited by a fixed proxy set.

Inspired by the compositionality of natural language, where a finite set of tokens supports rich expression, we represent collaborative information through the dynamic composition of reusable feature bases rather than through a single compressed feature or retrieved proxy. Building on this idea, we propose **ParaComm**, a learnable dictionary-driven framework for communication-efficient collaborative perception. ParaComm decomposes BEV features into locally maintained dictionary atoms and transmitted sparse continuous parameters. The dictionary atoms serve as reusable semantic bases, while the transmitted parameters are learned to capture observation-dependent variations over these bases. On the sender side, Joint Space-Atom Sparsification retains informative spatial locations and atom activations from the dense parameter map. Variable-Tile Parameter Sharing then reduces local redundancy among retained activations, and Quantized Representation converts the remaining parameters into low-precision messages. On the receiver side, collaborative features are explicitly reconstructed from the transmitted parameters using the shared dictionary. These features are then further refined through Communication Backbone Sharing, which improves feature quality while limiting additional model overhead before multi-agent fusion.

Compared with existing methods, ParaComm offers three main advantages: an efficient receiver pipeline, flexible communication control, and more interpretable representations. First, the core decoding step at the receiver reconstructs features via an explicit linear combination of the shared dictionary and sparse parameters, thereby avoiding a heavy feature decoder and keeping the receiver-side reconstruction lightweight and efficient. Second, ParaComm enables flexible control of the communication load through sender-side settings such as sparsity and local tile size, allowing different configurations to be explored without modifying the receiver-side reconstruction process or architecture. Finally, ParaComm explicitly decomposes collaborative information into shared semantic bases and observation-specific variations, making the communicated message more interpretable by establishing a clear correspondence among the transmitted parameters, the reconstruction process, and the recovered features.

We evaluate ParaComm on collaborative perception datasets spanning both simulated and real-world driving environments. The results demonstrate that ParaComm consistently achieves a stronger performance-communication trade-off than existing communication-efficient methods under comparable or lower communication budgets. Moreover, it also surpasses standard full-feature collaboration while transmitting only a small fraction of the feature payload. On OPV2V, for example, ParaComm reduces the communication volume from 34.375 MB to 99.688 KB, corresponding to a $353\times$ reduction, while improving the average detection performance by 0.277 percentage points over standard collaboration.

In summary, the main contributions are fourfold:

- We revisit what should be communicated in collaborative perception and introduce a Parameter-Centric Communication (PCC) perspective. PCC retains shared semantic bases locally and communicates compact parameters learned from BEV features to specify how these bases are combined for each observation.
- We propose ParaComm, a dictionary-driven framework for communication-efficient collaborative perception that learns reusable atoms shared across agents and conveys task-relevant information through sparse, observation-conditioned parameters.
- We develop Joint Space-Atom Sparsification, Variable-Tile Parameter Sharing, Quantized Representation, Dictionary-Based Explicit Reconstruction, and Communication Backbone Sharing, which jointly enable an efficient receiver-side pipeline, flexible communication control, and better representation interpretability.
- Extensive experiments on simulated and real-world datasets demonstrate that ParaComm consistently improves the performance-communication trade-off over existing communication-efficient methods and outperforms dense full-feature collaboration at substantially lower communication cost.

The rest of this paper is organized as follows. Section II reviews related work on communication-efficient collaborative perception and dictionary learning. Section III presents the problem formulation. Section IV introduces the proposed ParaComm framework. Section V provides extensive experimental results and analysis. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Communication-Efficient Collaborative Perception

Compared with single-agent perception [17]–[20], collaborative perception enhances the detection of occluded or distant targets by exchanging complementary information among multiple agents (e.g., vehicles and roadside units) [21]–[24]. However, collaboration incurs non-negligible communication overhead [25]–[28]. Existing collaborative perception methods are typically categorized by fusion stage into early, intermediate, and late collaboration [29]–[31]. Intermediate collaboration achieves a better balance between perception performance and communication cost, rendering it the predominant approach for collaborative perception [32]–[34]. Accordingly, this work focuses on intermediate collaboration.

In intermediate collaboration, many communication-efficient methods aim to reduce the cost of exchanging intermediate Bird's Eye View (BEV) features. Where2comm reduces spatial redundancy by learning importance maps and transmitting features from critical regions [13]. When2com [35] and Who2com [36] learn communication policies to determine informative interactions. ERMVP employs hierarchical feature sampling to extract representative feature vectors [37]. InfoCom filters features according to their task relevance, enabling effective collaboration under KB-level communication budgets [14]. How2comm [38] and GenComm [39] further improve communication efficiency through pragmatic communication design and compact conditional representations. These methods improve communication efficiency by selecting or compressing features. Nevertheless, their transmitted messages remain feature representations or feature surrogates during online collaboration, without explicitly separating reusable semantic bases from observation-specific variations. As a result, part of the communication budget may still be allocated to reusable content that could otherwise be retained locally.

Codebook-based methods compact online messages by mapping continuous features to codewords in a finite codebook shared across agents [40]. For example, CodeFilling transmits selected codeword indices [15], while PragComm further extends discrete indexing by mitigating spatiotemporal redundancy [16]. In these methods, the transmitted indices retrieve codewords that serve as predefined feature proxies. At the representation level, single-codeword lookup can be viewed as a special case of basis composition in which the parameters are restricted to one-hot vectors. ParaComm builds on local reuse but replaces discrete codeword selection with continuous parameters learned from BEV features. These parameters specify how multiple shared dictionary atoms are weighted and combined to form the current collaborative representation. Such parameter-driven combinations offer greater compositional flexibility for representing observation-specific variations over shared semantic bases. ParaComm further supports direct transmission of messages generated by the sender without requiring an additional request-response exchange.

B. Learnable Dictionary Learning

Dictionary learning aims to learn a set of dictionary atoms from data, enabling compact representation of signals as a sparse linear combination of a few atoms. Traditional dictionary learning employs alternating optimization to update both the dictionary and the sparse parameters [41]. While offering strong representational capacity, this approach poses challenges for integration with deep neural networks in end-to-end training and is often highly sensitive to initialization and hyperparameter choices. Recent work on learnable dictionaries has formulated dictionary updates and sparse coding as differentiable modules, enabling their seamless integration into deep learning frameworks. This approach has demonstrated strong performance in computer vision tasks, such as image denoising [42], super-resolution [43], and image generation [44].

Unlike prior work, this paper introduces learnable dictionaries to the domain of communication-efficient collaborative

perception. The shared dictionary atoms model the common semantic bases across agents, while sparse continuous parameters capture instance-specific variations arising from their observations. This design explicitly separates shared from differential information, transmitting only the latter during collaboration to maximize bandwidth efficiency. Given the sensitivity of collaborative perception to communication and computational constraints, this work enhances the learnable dictionary framework with several novel designs: Joint Space-Atom Sparsification, Variable-Tile Parameter Sharing, Dictionary-Based Explicit Reconstruction, and Communication Backbone Sharing. These adaptations tailor ParaComm to collaborative perception tasks under practical constraints.

III. PROBLEM FORMULATION

We consider a collaborative scenario with N agents, e.g., connected vehicles and roadside units equipped with perception capabilities. Each agent transmits information to others and incorporates received messages to overcome its perceptual limitations. This paper focuses on collaborative 3D object detection. Let $\mathcal{X}_i$ be the local sensor observation of the i -th agent, with corresponding annotations $\mathcal{Y}_i$. The online message from agent j to agent i is denoted by $\mathcal{M}_{j \rightarrow i}$, while $\Phi_\theta(\cdot)$ represents the collaborative detection model. Additionally, $g(\cdot, \cdot)$ is the detection performance metric, and $c(\cdot)$ denotes the communication cost of a message. In general, communication-efficient collaborative perception seeks a balance between detection performance and communication cost. A common formulation is to express this goal as maximizing overall detection performance under a total communication constraint:

$$\begin{aligned} \max_{\theta, \mathcal{M}} \quad & \sum_{i=1}^N g(\Phi_\theta(\mathcal{X}_i, \{\mathcal{M}_{j \rightarrow i}\}_{j \neq i}), \mathcal{Y}_i) \\ \text{s.t.} \quad & \sum_{i,j, j \neq i} c(\mathcal{M}_{j \rightarrow i}) \leq B, \end{aligned} \quad (1)$$

where B denotes a total communication constraint that characterizes the performance-communication trade-off. Existing methods typically design $\mathcal{M}_{j \rightarrow i}$ as intermediate features or their compressed representations, thereby optimizing the trade-off in BEV feature space.

IV. METHODOLOGY

A. Overview of ParaComm

Fig. 2 illustrates the overall framework of ParaComm. ParaComm maintains compatibility with existing collaborative perception systems, requiring no architectural modifications beyond the communication layer. It introduces parameterized message generation at the sender side and explicit reconstruction at the receiver side. This design achieves the transition from FCC to PCC by keeping reusable semantic bases local and communicating only the key sparse parameters that capture observation-specific variations. Specifically, each agent extracts an intermediate BEV feature using an encoder network and aligns it within a shared coordinate system. On the sender side, a lightweight shared Communication Backbone, along with Space and Atom Heads, transforms the BEV feature

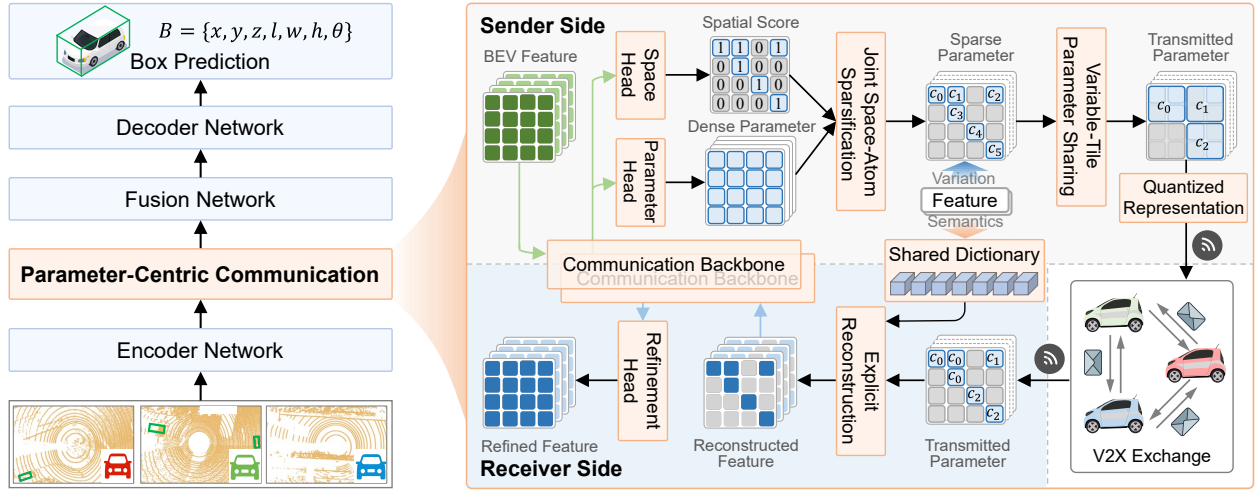


Fig. 2. Overall framework of ParaComm, a Parameter-Centric Communication pipeline for collaborative perception. On the sender side, a shared Communication Backbone, together with Space and Atom Heads, produces a spatial score and atom-level dense parameters. The spatial score is used in Joint Space-Atom Sparsification to sparsify the dense parameters, which are then further compacted by Variable-Tile Parameter Sharing and Quantized Representation to form the transmitted parameters. The resulting transmitted parameters are sent to the receiver, where they are explicitly reconstructed using locally shared dictionary atoms and further refined by a lightweight Refinement Head. The refinement stage reuses the Communication Backbone, reducing additional parameters and computation while improving feature quality.

into a spatial score and atom-level dense parameters. Joint Space-Atom Sparsification then processes these outputs, retaining only the most informative activations for collaboration. The resulting sparse parameters are further compressed via Variable-Tile Parameter Sharing and Quantized Representation before transmission. On the receiver side, the transmitted sparse parameters are explicitly reconstructed using locally shared dictionary atoms to recover collaborative features. These features are then refined to enhance representation quality. To further improve efficiency, ParaComm employs a Communication Backbone Sharing mechanism that reuses part of the feature extraction process. Finally, the refined collaborative features are fused with local features and fed into the decoder network for collaborative 3D object detection.

B. Parameter-Centric Communication Formulation

Building on the PCC perspective, ParaComm reformulates communication as transmitting observation-conditioned sparse parameters over a shared dictionary, rather than exchanging feature representations directly. Specifically, for the BEV feature $\mathbf{F}_j$ of sender agent j , ParaComm models $\mathbf{F}_j \approx \mathcal{R}(\mathbf{Z}_j, \mathbf{D})$, where $\mathbf{D}$ is a dictionary of reusable semantic bases, $\mathbf{Z}_j$ denotes the sender-specific sparse parameters, and $\mathcal{R}(\cdot)$ maps them to a feature-level approximation for collaboration. Here, $\mathbf{D}$ is learned jointly with the rest of the model parameters. Once learned, it is distributed to all agents together with the perception model and kept fixed during inference, avoiding online synchronization and per-round transmission cost. Consequently, the shared semantic bases are stored locally in $\mathbf{D}$, while agent-specific information is encoded in $\mathbf{Z}_j$.

Accordingly, ParaComm transmits only the sparse parameters, i.e., $\mathcal{M}_{j \rightarrow i} = \mathbf{Z}_j$. The receiver uses its local copy of $\mathbf{D}$ to reconstruct the collaborative feature as $\hat{\mathbf{F}}_j = \mathcal{R}(\mathbf{Z}_j, \mathbf{D})$, which is then fused with the receiver's local feature for detection. In this way, ParaComm replaces direct feature compression

with parameter transmission over a locally retained shared dictionary. Formally, the optimization objective of ParaComm can be expressed as

$$\begin{aligned} \max_{\theta, \mathbf{D}} \quad & \sum_{i=1}^N g\left(\Phi_{\theta}\left(\mathcal{X}_i, \{\hat{\mathbf{F}}_j\}_{j \neq i}\right), \mathcal{Y}_i\right) \\ \text{s.t.} \quad & \sum_{i, j, j \neq i} c(\mathbf{Z}_j) \leq B, \quad \hat{\mathbf{F}}_j = \mathcal{R}(\mathbf{Z}_j, \mathbf{D}). \end{aligned} \quad (2)$$

C. Sender-side Sparse Parameter Generation

1) *Communication Backbone and Parameter Encoding:* Let $\mathbf{F}_j \in \mathbb{R}^{C \times H \times W}$ denote the BEV feature of sender agent j , where $C \times H \times W$ is its feature shape. To better adapt this representation for communication modeling, we apply a lightweight Communication Backbone $\mathcal{B}(\cdot)$ that extracts contextual information and projects the feature into the latent communication space, yielding the communication feature:

$$\mathbf{F}_j^c = \mathcal{B}(\mathbf{F}_j) \in \mathbb{R}^{C \times H \times W}. \quad (3)$$

In contrast to stacking conventional convolutions, as shown in Fig. 3 (a), the Communication Backbone implements a local context modeling architecture inspired by the ConvNeXt block [45]. It consists of a depthwise convolution, a normalization layer, two 1×1 convolutions, a GELU activation, and a residual connection that preserves the original information. This design enhances the semantic expressiveness of the communication features at minimal computational cost while maintaining spatial resolution.

After acquiring the communication feature $\mathbf{F}_j^c$, we employ two parallel task heads to generate a dense parameter and a spatial score. The Parameter Head $\mathcal{H}_p(\cdot)$ produces the dense parameter map

$$\mathbf{S}_j = \mathcal{H}_p(\mathbf{F}_j^c) \in \mathbb{R}^{K \times H \times W}, \quad (4)$$

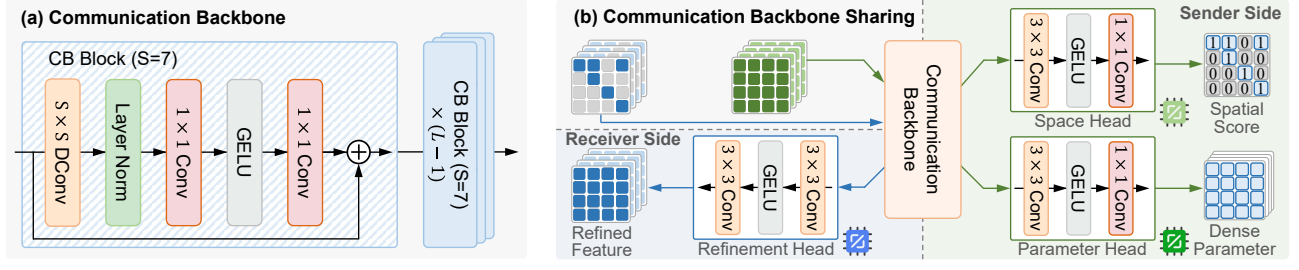


Fig. 3. Communication Backbone design in ParaComm. Left: the architecture of the Communication Backbone, which serves as a lightweight yet expressive feature extractor that maps BEV features into a latent communication space. Right: the illustration of the Communication Backbone Sharing, where the sender and receiver reuse the same backbone architecture and parameters, while attaching three task-specific heads with independent parameters. This design reduces computation and narrows the intrinsic feature gap between sender-side and receiver-side representations.

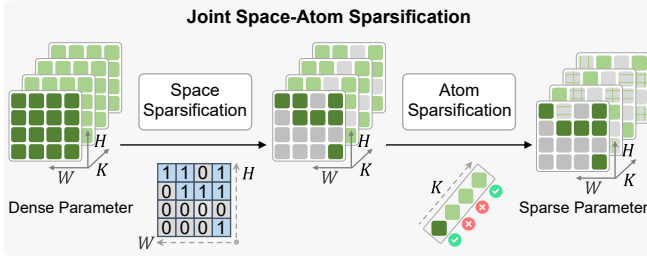


Fig. 4. Illustration of Joint Space-Atom Sparsification in ParaComm. Space Sparsification selects informative positions for communication, while Atom Sparsification retains only the most relevant dictionary activations.

where K represents the number of atoms in the shared dictionary, and $\mathbf{S}_j$ encodes the activation intensity of each spatial location across the dictionary atoms. Concurrently, the Space Head $\mathcal{H}_s(\cdot)$ generates a single-channel spatial score to quantify the relative contribution of each spatial location to collaborative communication, i.e.,

$$\mathbf{A}_j = \mathcal{H}_s(\mathbf{F}_j^c) \in \mathbb{R}^{1 \times H \times W}. \quad (5)$$

2) *Joint Space-Atom Sparsification*: The dense parameter $\mathbf{S}_j$ would incur substantial communication overhead if transmitted directly. To address this, we introduce Joint Space-Atom Sparsification for preserving only the essential parameters, as illustrated in Fig. 4. Unlike conventional dictionary learning that encourages sparsity mainly through regularization, ParaComm explicitly sparsifies the parameters over spatial locations and dictionary atoms. This is achieved by Space Sparsification, which selects the spatial locations for communication, and Atom Sparsification, which retains informative dictionary parameters at each selected location. Formally, this joint sparsification is expressed as

$$\tilde{\mathbf{S}}_j = \mathcal{T}_{\text{at}}(\mathbf{S}_j \odot \mathcal{T}_{\text{sp}}(\mathbf{A}_j)), \quad (6)$$

where $\mathcal{T}_{\text{sp}}(\cdot)$ and $\mathcal{T}_{\text{at}}(\cdot)$ denote the Space and Atom Sparsification operators, respectively. In the Space Sparsification step, the spatial score $\mathbf{A}_j$ is used to select informative locations among all $H \times W$ positions. The spatial mask is defined as $\mathbf{M}_j^{\text{sp}} = \mathcal{T}_{\text{sp}}(\mathbf{A}_j) \in \mathbb{R}^{1 \times H \times W}$, yielding the spatially filtered parameter map $\mathbf{S}_j^{\text{sp}} = \mathbf{S}_j \odot \mathbf{M}_j^{\text{sp}}$. Atom Sparsification is then applied independently to each retained spatial location. Specifically, for a location (h, w) , $\mathcal{T}_{\text{at}}$ processes the K -dimensional

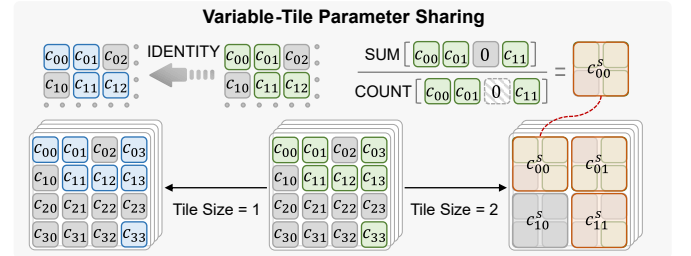


Fig. 5. Illustration of Variable-Tile Parameter Sharing in ParaComm. Non-zero sparse parameters are locally shared within spatial tiles to avoid repeatedly transmitting similar activations.

atom response vector $\mathbf{S}_j^{\text{sp}}(:, h, w)$ and keeps only informative activations. The resulting sparse parameter representation is denoted $\tilde{\mathbf{S}}_j = \mathcal{T}_{\text{at}}(\mathbf{S}_j^{\text{sp}})$. Here, the atom dimension indexes the dictionary atoms rather than the original feature channels, and each retained activation represents the parameter of a single dictionary atom used for subsequent reconstruction.

ParaComm supports three sparsification operators for both spatial and atom dimensions. Top- k Sparsification explicitly limits non-zero entries to the k largest activations and is formalized as $\mathcal{T}_{\text{topk}}(x; k) = x \odot M_{\text{topk}}(x; k)$, where $M_{\text{topk}}(x; k)$ is a binary mask that sets entries with the top k absolute values to 1 and the others to 0. Soft-threshold Sparsification is defined as $\mathcal{T}_{\text{soft}}(x; \lambda) = \text{sign}(x) \max(|x| - \lambda, 0)$, which suppresses responses with magnitudes below λ . ReLU-threshold Sparsification is given by $\mathcal{T}_{\text{relu}}(x; \tau) = \max(x - \tau, 0)$, which retains positive responses above τ . ParaComm adopts Top- k Sparsification as the default strategy, where preset numbers of spatial locations and atom activations are retained at each selected location. This yields a controllable upper bound on the number of non-zero parameters. The threshold-based operators serve as adaptive alternatives, allowing the retained parameter set to vary according to the response distribution.

3) *Variable-Tile Parameter Sharing*: While Joint Space-Atom Sparsification significantly reduces the number of non-zero parameters, the retained activations may still contain locally redundant parameter values. In particular, neighboring BEV locations often exhibit similar responses for the same dictionary atom. To exploit this local coherence, ParaComm introduces Variable-Tile Parameter Sharing, as depicted in Fig. 5. After the spatial-atom support is determined, this

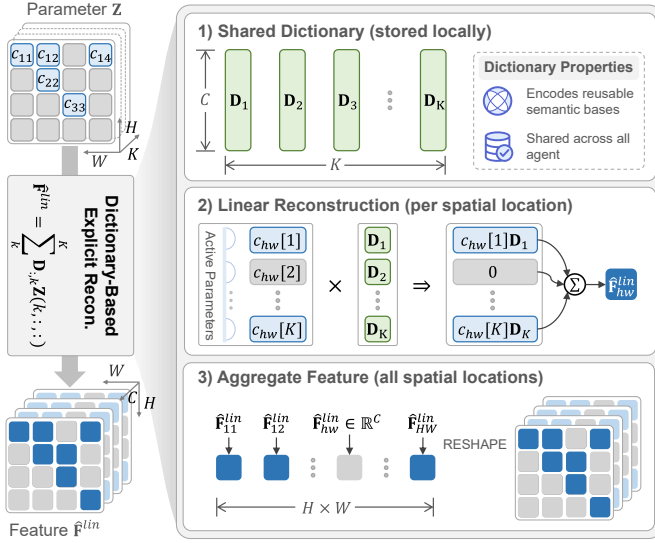


Fig. 6. Illustration of dictionary-based explicit reconstruction. The receiver keeps a shared dictionary locally and reconstructs the collaborative feature from the received sparse parameters.

step further compacts the sparse parameter map by sharing parameter values within local tiles. Specifically, we partition the sparse parameter map $\tilde{\mathbf{S}}_j \in \mathbb{R}^{K \times H \times W}$ for agent j into local tiles of size $T \times T$. For a given tile t , only the retained non-zero parameter responses within that tile are considered. For each atom k with non-zero activations in tile t , these responses are aggregated into a single shared parameter value:

$$\bar{s}_{j,t,k} = \frac{\sum_{(h,w) \in t} \tilde{\mathbf{S}}_j(k, h, w)}{\sum_{(h,w) \in t} \mathbb{I}(\tilde{\mathbf{S}}_j(k, h, w) \neq 0)}, \quad (7)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. Thereafter, all non-zero positions of atom k within tile t share the value $\bar{s}_{j,t,k}$, while originally zero positions remain unchanged. During decoding, the shared value is restored to its retained positions at the receiver. The tile size T governs the trade-off between detail preservation and communication reduction, letting ParaComm adapt to different bandwidth budgets without changing the receiver architecture. Even under shared parameters, neighboring positions can recover discriminative representations through distinct combinations of dictionary atoms.

4) *Quantized Sparse Parameter Representation*: After parameter sparsification and local sharing, the message is already sparse and compact, but its values are still represented in standard 32-bit floating-point format. In ParaComm, once the spatial locations, atom activations, and tile-sharing pattern have been determined, the remaining values mainly control the relative contribution of retained dictionary atoms in linear reconstruction. Thus, full-precision parameters are unnecessary, and we map the continuous sparse parameters into finite-bit representations to reduce numerical overhead. Specifically, $\tilde{\mathbf{S}}_j$ is quantized using symmetric uniform quantization:

$$\hat{\mathbf{S}}_j = \mathcal{Q}(\tilde{\mathbf{S}}_j; b_q), \quad (8)$$

where $\mathcal{Q}(\cdot; b_q)$ denotes the quantization operator and b_q is the bit-width. In practice, quantization is performed using

$$\mathcal{Q}(x; b_q) = \alpha \cdot \frac{\text{Round}(\text{Clip}(x/\alpha, -1, 1) \cdot q_{\max})}{q_{\max}}, \quad (9)$$

where $\alpha = \max |x|$ and $q_{\max} = 2^{b_q-1} - 1$. This design adapts to the dynamic range of retained parameters under a limited bit budget, and straight-through estimation during training enables end-to-end optimization. The transmitted message $\mathbf{Z}_j$ then consists of the quantized parameter values together with the structural metadata required for decoding, including retained spatial locations, atom indices, and tile-sharing assignments.

D. Receiver-side Feature Reconstruction

1) *Dictionary-Based Explicit Reconstruction*: As illustrated in Fig. 6, the receiver-side decoding process of ParaComm reconstructs intermediate features from the received sparse parameters using a dictionary, rather than directly generating them with a complex decoder. This design preserves a clear structural correspondence between the communicated parameters and the recovered features, thereby distinguishing ParaComm from the implicit feature recovery of existing methods. Let $\mathbf{D} \in \mathbb{R}^{C \times K}$ denote the shared dictionary consisting of K atoms, where the k -th atom is a C -dimensional semantic basis vector. At each spatial location, the receiver reconstructs a feature response by taking a linear combination of the dictionary atoms with the parameters as weights. These reconstructed responses are then aggregated across all spatial locations to form the intermediate feature map. This linear reconstruction is expressed as follows:

$$\hat{\mathbf{F}}_j^{\text{lin}} = \mathcal{R}(\mathbf{Z}_j, \mathbf{D}) = \sum_{k=1}^K \mathbf{D}_{:,k} \mathbf{Z}_j(k, :, :), \quad (10)$$

where $\hat{\mathbf{F}}_j^{\text{lin}} \in \mathbb{R}^{C \times H \times W}$ is the reconstructed feature. Because $\mathbf{Z}_j$ is sparse, only the atom activations corresponding to the few non-zero parameters contribute to the above aggregation. This process can also be rewritten as follows, with the matrix multiplication applied independently at each spatial location:

$$\hat{\mathbf{F}}_j^{\text{lin}} = \mathbf{D} \mathbf{Z}_j. \quad (11)$$

This reconstruction process reflects the key idea of ParaComm: the shared dictionary $\mathbf{D}$ provides reusable semantic bases across agents, while the communicated message $\mathbf{Z}_j$ encodes only the observation-specific variations beyond those bases. Instead of a complex neural decoder, the receiver reconstructs intermediate features by linearly combining the received sparse parameters with the locally stored shared dictionary, keeping reconstruction simple and efficient while improving interpretability through a clearer correspondence between the message and the recovered features.

2) *Refinement with Communication Backbone Sharing*: Although explicit linear reconstruction can recover the primary structure of intermediate features, it may not sufficiently capture fine local details or inter-channel dependencies. To further unleash the representational power, ParaComm adds a lightweight refinement branch after explicit reconstruction.

As shown in Fig. 3(b), ParaComm adopts a Communication Backbone Sharing strategy, where the sender-side Communication Backbone is reused for receiver-side refinement instead of introducing a separate refinement backbone. Specifically, the receiver projects $\hat{\mathbf{F}}_j^{lin}$ to the required input dimension and processes it through the shared Communication Backbone $\mathcal{B}(\cdot)$ followed by a Refinement Head $\mathcal{H}_r(\cdot)$ for feature refinement. The final refined feature map is then obtained as:

$$\hat{\mathbf{F}}_j = \mathcal{H}_r(\mathcal{B}(\hat{\mathbf{F}}_j^{lin})) \in \mathbb{R}^{C \times H \times W}. \quad (12)$$

Reusing the backbone yields two benefits. First, it eliminates the need for a separate large-scale feature extraction module at the receiver, which reduces the number of model parameters. Second, by reusing the same feature transformation space at both the sender and receiver, it helps reduce the distribution gap between the communicated parameters and the recovered features, which in turn improves reconstruction stability.

After refinement, receiver i fuses its local feature $\mathbf{F}_i$ with the refined features $\{\hat{\mathbf{F}}_j \mid j \in \{1, \dots, N\}, j \neq i\}$ and then feeds the resulting representation into the detection head for collaborative 3D object detection.

E. Training Objective

ParaComm is trained with three objectives: an object detection loss, a feature reconstruction loss, and a dictionary consistency loss. First, the object detection loss $\mathcal{L}_{\text{det}}$, composed of standard classification and regression losses [12], [46], optimizes the collaborative perception task and helps preserve task-relevant information in the transmitted parameters. Second, the feature reconstruction loss $\mathcal{L}_{\text{rec}} = \sum_{j=1}^N \ell_{\text{rec}}(\hat{\mathbf{F}}_j, \mathbf{F}_j)$ enforces the recovery of intermediate features, where $\ell_{\text{rec}}(\cdot, \cdot)$ is implemented with either L1 loss or MSE loss. Since $\hat{\mathbf{F}}_j$ is reconstructed from the transmitted parameters $\mathbf{Z}_j$ and the shared dictionary $\mathbf{D}$, this loss further promotes the retention of sender-dependent information required for BEV feature recovery. Finally, the dictionary consistency loss $\mathcal{L}_{\text{con}} = \|\tilde{\mathbf{D}}^\top \tilde{\mathbf{D}} - \mathbf{I}\|_F^2$ reduces redundancy among dictionary atoms, where $\tilde{\mathbf{D}}$ denotes the L_2 -normalized dictionary, $\mathbf{I}$ is the identity matrix, and $\|\cdot\|_F$ denotes the Frobenius norm. This term discourages highly correlated atoms and promotes diverse semantic bases for stable dictionary-based reconstruction. Accordingly, the overall training objective is defined as

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}, \quad (13)$$

where λ_{rec} and λ_{con} control the relative contributions of the reconstruction and consistency terms. Joint optimization encourages effective collaborative detection, accurate feature recovery, and a well-conditioned shared dictionary.

V. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: Following prior work [14], we evaluate the proposed method on three widely used collaborative perception datasets covering simulated and real-world environments, as well as Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2X) scenarios. Specifically, OPV2V is a large-scale V2V dataset constructed using the CARLA simulator

[12]. It provides synchronized multi-view observations with 3D bounding box annotations and is primarily used for benchmarking methods in multi-vehicle settings. In contrast, V2XSet is also a large-scale simulated dataset but incorporates data from RSUs, making it suitable for evaluating V2X collaborative perception systems [47]. DAIR-V2X is a more challenging real-world V2X dataset, comprising real data synchronously captured by vehicle-mounted and roadside sensors [48]. Since the original annotations lack complete 360° coverage, we employ its extended annotation version [46]. Unless otherwise specified, all experiments follow the default data splits and preprocessing configurations for each dataset.

2) *Evaluation Metrics*: Our primary evaluation metric is Average Precision (AP). We report AP at Intersection over Union (IoU) thresholds of 0.30, 0.50, and 0.70, denoted AP@30, AP@50, and AP@70. Following prior work [14], [15], we measure communication cost as the actual transmitted volume per collaborative round in MB or KB, rather than normalizing all methods to a fixed common budget. For all methods, the reported volume is measured from the full transmitted message, including both payload values and required auxiliary information. For ParaComm, this includes quantized parameter values together with retained spatial locations, atom indices, and other decoding metadata. For runtime efficiency, we measure the end-to-end inference latency of each method and report the results in milliseconds.

3) *Baselines*: To evaluate ParaComm, we compare it with No Communication as a non-collaborative reference, Late Collaboration as a low-cost result-level baseline, and Standard Collaboration as a full-feature baseline without communication compression. We also compare ParaComm with several communication-efficient baselines, including Where2comm [13], ERMVP [37], GenComm [39], InfoCom [14], and PragComm [16]. For consistency with our per-round communication setting, PragComm is evaluated without inter-frame redundancy reduction. We also implement PragComm*, a variant configured to operate at a communication volume comparable to ParaComm for a more direct codebook-based comparison. Following prior work [14], we evaluate all methods under a unified collaborative perception setup with the same fusion mechanism. We use CoAlign as the default base model [46] and further evaluate generalizability on two alternative frameworks, AttFuse [12] and MKD-Cooper [49].

4) *Implementation Details*: ParaComm is implemented in PyTorch, and all experiments are conducted on a single NVIDIA RTX 4090 GPU. The Communication Backbone contains two basic blocks by default to balance representation capacity and parameter efficiency. The dictionary contains $K = 64$ atoms and is initialized from a normal distribution. It is optimized jointly with the rest of the model during training, and shared across all agents during inference. Joint Space-Atom Sparsification uses the Top- k operator by default because it provides direct control over the communication budget. Specifically, we retain the top 10% of the $H \times W$ spatial positions and the top 4 atom activations. The default quantization bit-width b_q is 4 bits. The tile size T defaults to 1, with the optional set $\{1, 2, 4, 8\}$. The loss weights λ_{rec} and λ_{con} are set to 0.2 and 0.001, respectively. We adopt a

TABLE I

QUANTITATIVE COMPARISON ON THREE REPRESENTATIVE DATASETS. ALL COMMUNICATION-EFFICIENT METHODS ARE IMPLEMENTED USING THE SAME COALIGN BACKBONE FOR FAIR COMPARISON. PARACOMM ACHIEVES THE BEST OVERALL PERFORMANCE AMONG COMMUNICATION-EFFICIENT METHODS. AP VALUES ARE PRESENTED AS MEAN $\pm$ STANDARD DEVIATION. FOR CLARITY, THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN LIGHT GREEN AND LIGHT GRAY, RESPECTIVELY.

Dataset	Communication Method	Publication	Comm. Volume	AP@30	AP@50	AP@70
OPV2V [12]	No Communication	/	0	0.8657 ± 0.0029	0.8470 ± 0.0015	0.7274 ± 0.0051
	Late Collaboration	/	7.031 KB	0.9627 ± 0.0014	0.9491 ± 0.0028	0.8506 ± 0.0030
	Standard Collaboration [46]	ICRA 2023	34.375 MB	0.9705 ± 0.0018	0.9656 ± 0.0012	0.9221 ± 0.0024
	Where2comm [13]	NeurIPS 2022	3.468 / 3.437 MB	0.9539 ± 0.0032	0.9469 ± 0.0025	0.8805 ± 0.0055
	ERMVP [37]	CVPR 2024	0.771 / 0.688 MB	0.9624 ± 0.0026	0.9550 ± 0.0031	0.9136 ± 0.0046
	GenComm [39]	NeurIPS 2025	1.074 / 1.074 MB	0.9713 ± 0.0013	0.9646 ± 0.0017	0.9072 ± 0.0032
	PragComm [16]	TPAMI 2026	0.732 / 0.671 MB	0.9724 ± 0.0016	0.9659 ± 0.0030	0.9153 ± 0.0025
	PragComm* [16]	TPAMI 2026	89.375 / 27.500 KB	0.9663 ± 0.0030	0.9618 ± 0.0027	0.9087 ± 0.0052
V2XSet [47]	InfoCom [14]	AAAI 2026	38.813 / 7.875 KB	0.9707 ± 0.0023	0.9662 ± 0.0026	0.9208 ± 0.0022
	ParaComm (Ours)	/	99.688 / 27.500 KB	0.9732 ± 0.0005	0.9672 ± 0.0010	0.9261 ± 0.0028
	No Communication	/	0	0.8025 ± 0.0027	0.7728 ± 0.0032	0.6209 ± 0.0049
	Late Collaboration	/	7.031 KB	0.9327 ± 0.0016	0.9069 ± 0.0024	0.7135 ± 0.0054
	Standard Collaboration [46]	ICRA 2023	34.375 MB	0.9312 ± 0.0025	0.9216 ± 0.0013	0.8418 ± 0.0031
	Where2comm [13]	NeurIPS 2022	3.468 / 3.437 MB	0.8844 ± 0.0033	0.8595 ± 0.0029	0.7401 ± 0.0056
	ERMVP [37]	CVPR 2024	0.771 / 0.688 MB	OOM	OOM	OOM
	GenComm [39]	NeurIPS 2025	1.074 / 1.074 MB	0.9352 ± 0.0012	0.9241 ± 0.0018	0.8451 ± 0.0026
DAIR-V2X [48]	PragComm [16]	TPAMI 2026	0.732 / 0.671 MB	0.9371 ± 0.0024	0.9257 ± 0.0015	0.8394 ± 0.0047
	PragComm* [16]	TPAMI 2026	89.375 / 27.500 KB	0.9331 ± 0.0031	0.9213 ± 0.0026	0.8236 ± 0.0050
	InfoCom [14]	AAAI 2026	38.813 / 7.875 KB	0.9364 ± 0.0015	0.9270 ± 0.0011	0.8483 ± 0.0023
	ParaComm (Ours)	/	99.688 / 27.500 KB	0.9403 ± 0.0014	0.9292 ± 0.0004	0.8513 ± 0.0012
	No Communication	/	0	0.7286 ± 0.0030	0.6838 ± 0.0025	0.5679 ± 0.0053
	Late Collaboration	/	7.031 KB	0.7987 ± 0.0017	0.6718 ± 0.0033	0.4693 ± 0.0055
	Standard Collaboration [46]	ICRA 2023	24.609 MB	0.8298 ± 0.0013	0.7838 ± 0.0016	0.6361 ± 0.0027
	Where2comm [13]	NeurIPS 2022	2.481 / 2.461 MB	0.8039 ± 0.0032	0.7547 ± 0.0028	0.6058 ± 0.0048
	ERMVP [37]	CVPR 2024	0.551 / 0.492 MB	0.8223 ± 0.0024	0.7787 ± 0.0014	0.6331 ± 0.0025
	GenComm [39]	NeurIPS 2025	0.769 / 0.769 MB	0.8103 ± 0.0015	0.7660 ± 0.0027	0.6225 ± 0.0044
	PragComm [16]	TPAMI 2026	0.522 / 0.481 MB	0.8254 ± 0.0012	0.7859 ± 0.0013	0.6405 ± 0.0021
	PragComm* [16]	TPAMI 2026	61.523 / 19.688 KB	0.8082 ± 0.0029	0.7656 ± 0.0031	0.6270 ± 0.0052
	InfoCom [14]	AAAI 2026	26.840 / 5.922 KB	0.8231 ± 0.0018	0.7794 ± 0.0023	0.6380 ± 0.0030
	ParaComm (Ours)	/	70.137 / 19.688 KB	0.8282 ± 0.0012	0.7873 ± 0.0011	0.6499 ± 0.0030

TABLE II

QUANTITATIVE EVALUATION OF COMMUNICATION-EFFICIENT METHODS WITH ALTERNATIVE COLLABORATIVE PERCEPTION BACKBONES ON THE OPV2V DATASET. THE RESULTS SHOW THAT PARACOMM GENERALIZES EFFECTIVELY ACROSS DIFFERENT BACKBONES. AP VALUES ARE PRESENTED AS MEAN $\pm$ STANDARD DEVIATION. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN LIGHT GREEN AND LIGHT GRAY, RESPECTIVELY.

AttFuse [12]					MKD-Cooper [49]		
Communication Method	Comm. Volume	AP@30	AP@50	AP@70	AP@30	AP@50	AP@70
Standard Collaboration	8.594 MB	0.9549 ± 0.0028	0.9322 ± 0.0014	0.8048 ± 0.0052	0.9571 ± 0.0015	0.9364 ± 0.0031	0.8146 ± 0.0026
Where2comm [13]	0.866 / 0.859 MB	0.9166 ± 0.0032	0.9009 ± 0.0027	0.8011 ± 0.0056	0.9140 ± 0.0026	0.8967 ± 0.0013	0.7968 ± 0.0045
ERMVP [37]	0.192 / 0.172 MB	0.9218 ± 0.0013	0.9033 ± 0.0030	0.7930 ± 0.0024	0.9107 ± 0.0033	0.8952 ± 0.0028	0.7838 ± 0.0054
GenComm [39]	0.269 / 0.269 MB	0.9588 ± 0.0012	0.9391 ± 0.0016	0.8205 ± 0.0022	0.9582 ± 0.0025	0.9370 ± 0.0011	0.8167 ± 0.0048
PragComm [16]	0.181 / 0.168 MB	0.9592 ± 0.0029	0.9388 ± 0.0012	0.8181 ± 0.0047	0.9563 ± 0.0014	0.9354 ± 0.0026	0.8204 ± 0.0027
PragComm* [16]	20.625 / 6.875 KB	0.9533 ± 0.0031	0.9349 ± 0.0025	0.8218 ± 0.0051	0.9546 ± 0.0027	0.9324 ± 0.0032	0.8127 ± 0.0030
InfoCom [14]	9.593 / 2.719 KB	0.9579 ± 0.0016	0.9356 ± 0.0024	0.8162 ± 0.0025	0.9641 ± 0.0011	0.9485 ± 0.0015	0.8338 ± 0.0021
ParaComm (Ours)	24.063 / 6.875 KB	0.9610 ± 0.0005	0.9415 ± 0.0011	0.8283 ± 0.0012	0.9636 ± 0.0014	0.9413 ± 0.0013	0.8275 ± 0.0030

communication warm-up strategy during training. The base collaborative model is first trained for 30 epochs without the ParaComm communication module. ParaComm is then integrated and the full model is trained for another 20 epochs. The training batch size is 4, with an initial learning rate of 0.002 that decays at predefined milestones. A batch size of 1 is used during inference to avoid variance from batch processing.

B. Quantitative Evaluation

Table I compares communication-efficient methods on three representative datasets using the same CoAlign backbone,

with No Communication, Late Collaboration, and Standard Collaboration as references. In the Communication Volume column, black values represent the total message size, whereas gray values denote only the main payload size. We observe several consistent trends. First, all collaborative methods outperform the non-collaborative baseline, indicating that inter-agent communication helps alleviate the inherent limitations of single-agent perception. Second, compared with Late Collaboration, feature-level collaboration generally achieves stronger performance, highlighting the value of richer semantic interaction across agents. Finally, several communication-efficient

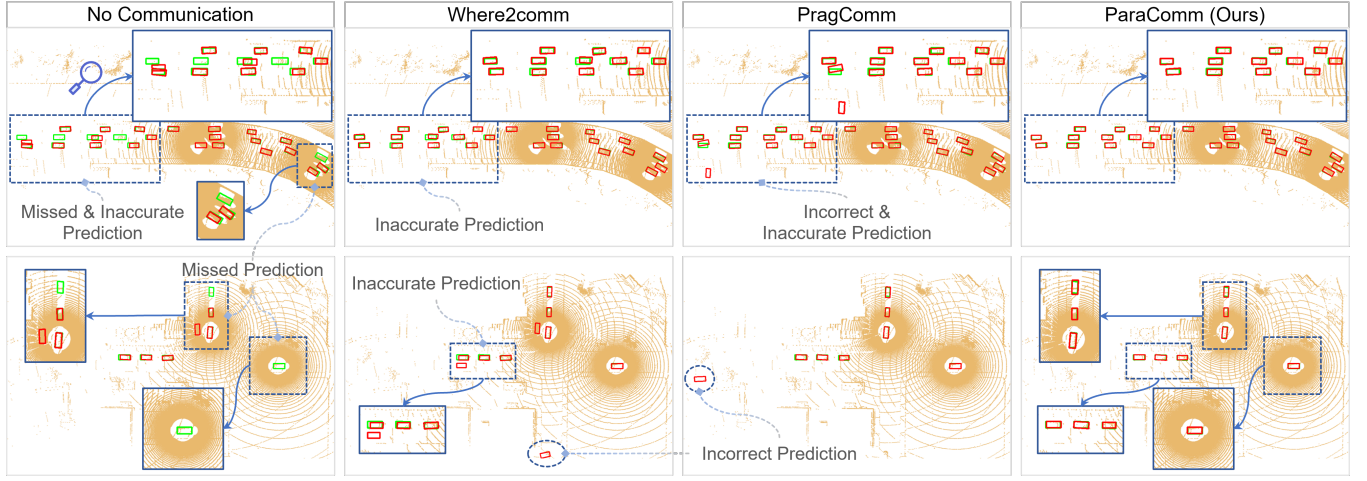


Fig. 7. Qualitative comparison between ParaComm and representative baselines. Blue dashed boxes highlight challenging regions, and zoomed-in views are provided for clearer comparison. While communication generally improves over No Communication, ParaComm yields more complete detections and more accurate localization than other communication-efficient methods, especially for distant, occluded, and densely distributed objects.

TABLE III

COMMUNICATION VOLUME UNDER DIFFERENT PARACOMM DESIGN CHOICES. RESULTS ARE REPORTED AS COMMUNICATION VOLUME AND REDUCTION RATIO RELATIVE TO THE FULL FEATURE TRANSMISSION.

Design	Configuration	Comm. Volume	Reduction
Baseline	-	34.375 MB	0
Joint Space-Atom Sparsification	Space Top-10%	3.468 / 3.437 MB	$\downarrow 9.9\times$
	Atom Top-4	2.551 / 2.148 MB	$\downarrow 13.5\times$
	Both	0.285 / 0.215 MB	$\downarrow 120.6\times$
Variable-Tile Parameter Sharing	Tile Size $T=2$	1.779 / 1.719 MB	$\downarrow 19.3\times$
	Tile Size $T=4$	0.490 / 0.430 MB	$\downarrow 70.2\times$
Quantized Representation	Bit-Width $b_q = 4$	4.297 / 4.297 MB	$\downarrow 8.0\times$
ParaComm (Full)	-	99.688 / 27.500 KB	$\downarrow 353.1\times$

baselines still underperform full-feature collaboration, suggesting that reducing communication alone is insufficient without preserving critical collaborative information.

Compared with Standard Collaboration, ParaComm maintains dramatically lower communication cost while achieving superior detection performance. For example, it reduces communication by up to $359\times$ and improves AP@70 on DAIR-V2X by 2.2% over Standard Collaboration. This is likely because ParaComm reconstructs collaborative features from compact sparse parameters rather than directly transmitting full feature content. Such a representation can better preserve collaboration-critical information while avoiding redundant or noisy content. *Compared with existing communication-efficient methods, ParaComm achieves a favorable performance-communication trade-off.* Although InfoCom is a strong low-bandwidth baseline, ParaComm keeps the communication cost below 100 KB per collaborative round while achieving better overall performance. On the real-world DAIR-V2X dataset, ParaComm improves AP@70 over InfoCom by 1.2 percentage points, suggesting that directly allocating communication to observation-specific parameters is more robust than selecting or compressing features in real-

TABLE IV

COMPARISON OF DIFFERENT TRAINING STRATEGIES FOR PARACOMM. THE PROPOSED COMMUNICATION WARM-UP CONSISTENTLY OUTPERFORMS BOTH TRAINING FROM SCRATCH AND FINE-TUNING.

Training Strategy	AP@30	AP@50	AP@70
Training from Scratch	0.9712	0.9646	0.8951
Fine-Tuning	0.9699	0.9641	0.9195
Communication Warm-up	0.9732	0.9672	0.9261

world sensing scenarios. PragComm and PragComm* are the most relevant codebook-based baselines. On V2XSet, ParaComm achieves a $7.5\times$ reduction in communication volume relative to PragComm while improving AP@70 by 1.42%, and it further surpasses PragComm* by up to 3.36% AP@70 under a comparable communication volume. ParaComm's improvements are therefore attributable not only to a simpler protocol, but also to the higher representational efficiency of combining multiple atoms compared to discrete codeword selection.

We further evaluate ParaComm using two alternative collaborative backbones, AttFuse and MKD-Cooper, as summarized in Table II. When integrated into AttFuse, ParaComm consistently achieves the best performance across all evaluated IoU thresholds, demonstrating its effectiveness beyond the primary backbone. When integrated into MKD-Cooper, ParaComm remains highly competitive, ranking second overall and outperforming most communication-efficient baselines. However, InfoCom offers a better accuracy-communication trade-off on this backbone, achieving 0.8338 AP@70 with 9.593 KB of communication, compared with 0.8275 AP@70 and 24.063 KB for ParaComm. This difference may arise from the interaction between the communication and the fusion mechanism. AttFuse fuses features from different agents at corresponding spatial locations. ParaComm represents reusable patterns by combining a small number of atoms, thereby reducing transmission overhead while preserving complementary information. By contrast, MKD-Cooper explicitly derives fusion weights from channel and spatial responses. Sparsification and

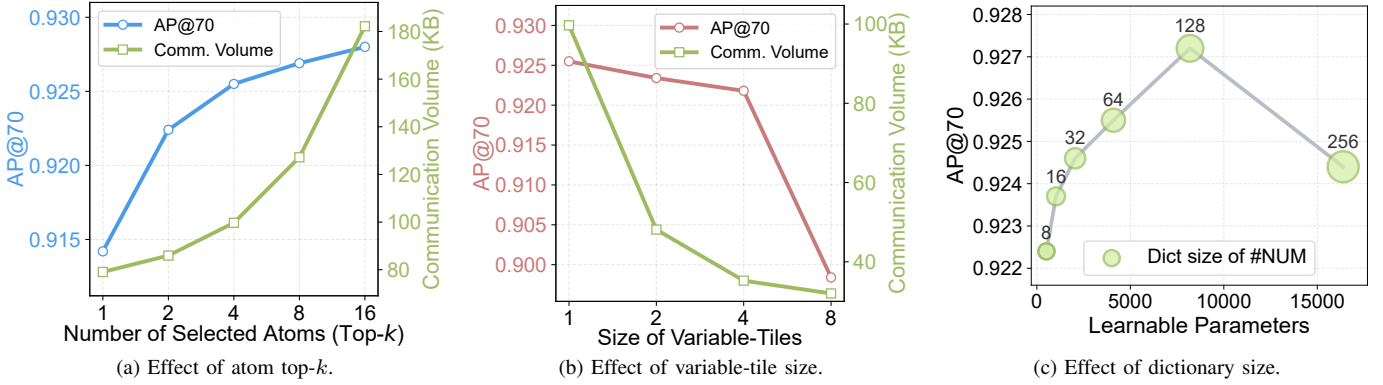


Fig. 8. Trade-off analysis of key design choices in ParaComm. Different design configurations entail different communication volumes or parameter counts, which in turn lead to varying performance–communication trade-offs.

quantization may alter both the feature content and the resulting fusion weights, limiting the performance–communication trade-offs of the compositional representation.

C. Qualitative Evaluation

Fig. 7 presents qualitative comparisons between ParaComm and representative baselines in challenging scenarios. Compared with the No Communication baseline, communication-enabled methods generally recover more objects that are difficult for a single agent to detect, particularly under conditions of long range, occlusion, or limited field of view, illustrating the benefit of inter-agent information sharing. Nevertheless, clear differences remain among communication-efficient methods. Where2comm improves recall in several highlighted regions, but also introduces occasional false positives in ambiguous background areas. PragComm shows weaker recovery in crowded scenes, with less accurate localization for some closely spaced vehicles. In contrast, ParaComm recovers more difficult objects in the highlighted regions, including distant, partially visible, and closely packed vehicles. Its predictions also exhibit tighter boxes and clearer separation between adjacent targets. These observations are consistent with the quantitative results and suggest that ParaComm preserves more useful collaboration cues under constrained communication.

D. Deeper Analysis

1) *Effect of Different Designs on Communication Volume:* Tab. III shows how different design choices affect the final communication volume. Each component yields a clear reduction relative to the baseline transmission volume of 34.375 MB. Among them, Joint Space-Atom Sparsification provides the largest single-stage gain. When Space and Atom Sparsification are applied together, the communication volume drops to 0.285 MB. This highlights the importance of jointly removing spatial redundancy and unnecessary atom activations. Variable-Tile Parameter Sharing further reduces communication by sharing locally retained activations, with larger tile sizes yielding stronger compression. For instance, using $T = 4$ reduces the communication volume to 0.490 MB. Quantized Representation lowers sparse-parameter transmission cost from the value-precision perspective, achieving an

$8.0\times$ reduction with 4-bit quantization. When all components are combined, ParaComm (Full) reduces the communication volume to only 99.688 KB, yielding an overall $353\times$ reduction. These results show that ParaComm achieves its high communication efficiency through the complementary effects of sparsification, local sharing, and low-precision representation.

2) *Effect of Communication Warm-up Strategy:* Tab. IV compares three training strategies for ParaComm, namely Training from Scratch, Fine-Tuning, and the proposed Communication Warm-up. Among them, Communication Warm-up consistently achieves the best results across all metrics, reaching 0.9732 AP@30, 0.9672 AP@50, and 0.9261 AP@70. Compared with Training from Scratch, these results indicate that jointly learning the collaborative perception backbone and the parameter communication module from random initialization is more challenging. The improvement over Fine-Tuning, which updates only the ParaComm-related parameters while freezing the remaining backbone, indicates that adapting the communication module alone is insufficient to fully align sparse parameter transmission with the downstream collaborative detection objective. In contrast, the proposed Communication Warm-up first establishes a well-trained collaborative perception backbone and then introduces Parameter-Centric Communication, leading to a more stable optimization process and better final performance.

3) *Effect of the Number of Selected Atoms:* Fig. 8a analyzes the effect of different atom top- k values on both detection performance and communication volume. We report AP@70 here because it imposes the strictest localization requirement. As the number of selected atoms increases, AP@70 consistently improves, indicating that retaining more atom activations provides stronger representation capacity for the transmitted sparse parameters. However, this improvement gradually saturates, while the communication volume grows much faster. In particular, increasing k from very small values to a moderate range improves detection performance, whereas further increases bring only marginal benefit along with higher communication cost. Even at $k = 1$, ParaComm still differs from Codebook by using a continuous parameter rather than a discrete codeword index, which preserves greater expressiveness under this sparse setting. Overall, a moderate

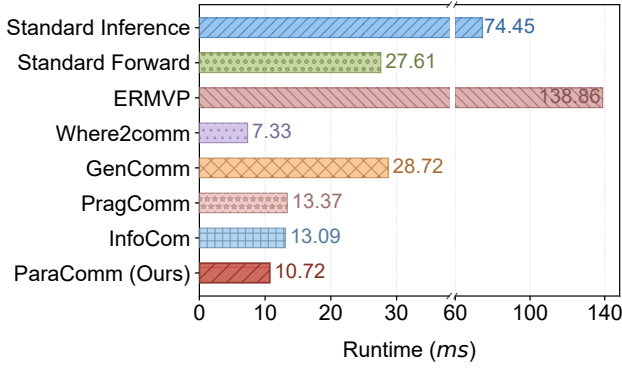


Fig. 9. Additional inference latency of communication-efficient methods, with Standard Forward and Standard Inference as references. ParaComm is a computationally efficient communication solution with low runtime overhead.

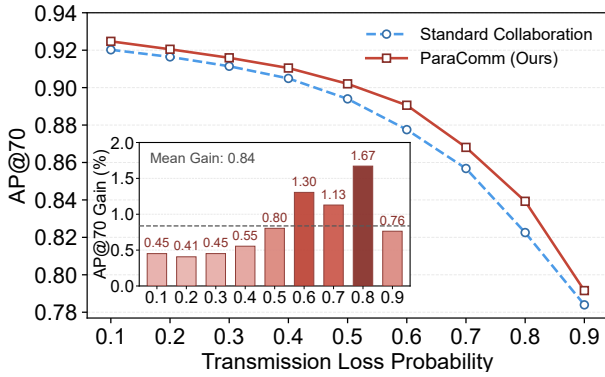


Fig. 10. AP@70 performance and absolute gain of ParaComm over Standard Collaboration under varying transmission loss probabilities. ParaComm achieves more reliable detection under communication loss.

atom top- k appears to offer a more suitable balance between communication efficiency and representational capacity.

4) *Effect of Variable-Tile Size*: Fig. 8b analyzes the effect of different variable-tile sizes on both AP@70 and communication volume. Larger tile sizes substantially reduce communication volume, indicating that broader local sharing is effective in compressing sparse parameter messages. This reduction, however, is achieved by enforcing stronger sharing within each local region, which weakens the ability of the transmitted representation to preserve fine-grained instance-specific details. Consequently, detection accuracy decreases as the tile size grows. In practice, moderate tile sizes provide a better balance, as they already achieve most of the communication savings while retaining sufficient local variation for accurate collaborative perception. By contrast, excessively large tiles make the local approximation overly coarse and therefore lead to a more pronounced performance drop. This indicates that tile size should be selected to balance communication reduction with the preservation of informative local variations.

5) *Effect of Dictionary Size*: Fig. 8c shows the effect of dictionary size on detection performance, with bubble size indicating the corresponding number of learnable parameters. As the dictionary becomes larger, AP@70 first improves, showing that a richer set of dictionary atoms helps better capture reusable semantic bases and supports more expressive parameter reconstruction. However, this trend does not

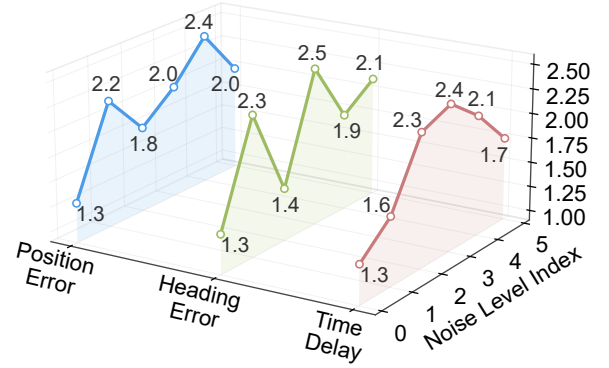


Fig. 11. Average absolute AP gain of ParaComm over Standard Collaboration under multiple communication perturbations. ParaComm retains positive gains throughout the evaluated range.

TABLE V
CROSS-DOMAIN EVALUATION ON THE CULVER CITY DIGITAL TOWN SUBSET. PARACOMM ACHIEVES STRONG COLLABORATIVE PERCEPTION PERFORMANCE IN MORE CHALLENGING SCENES.

Communication Method	AP@30	AP@50	AP@70	Delta (%)
Standard Collaboration	0.8948	0.8774	0.7965	/
Where2comm	0.8658	0.8432	0.7358	-4.13
ERMVP	0.8599	0.8483	0.7791	-2.71
GenComm	0.8964	0.8768	0.7671	-0.95
PragComm	0.8931	0.8784	0.7850	-0.41
PragComm*	0.8940	0.8784	0.7815	-0.49
InfoCom	0.8860	0.8711	0.7845	-0.90
ParaComm (Ours)	0.9023	0.8869	0.8017	+0.74

continue indefinitely. Once the dictionary size becomes too large, the performance starts to decline, suggesting that excessive dictionary capacity introduces additional redundancy and makes optimization less effective. In other words, a dictionary that is too small limits representation power, while an overly large one reduces parameter efficiency and may degrade the quality of the learned shared semantics. These results indicate that a relatively small dictionary is sufficient to capture the reusable semantic basis needed for collaboration without introducing unnecessary complexity.

6) *Inference Latency Analysis*: Fig. 9 compares the additional runtime overhead introduced by different communication-efficient methods, with Standard Forward and Standard Inference shown as references. The results indicate that lower communication volume does not necessarily translate into lower processing overhead, since some methods incur expensive message generation, reconstruction, or multi-round interaction. Among all communication-efficient baselines, Where2comm introduces the smallest overhead, while ParaComm ranks second with only 10.72 ms, remaining lower than InfoCom and PragComm, and substantially below GenComm and ERMVP. Notably, ERMVP introduces even higher overhead than Standard Inference, making communication efficiency alone inadequate for overall acceleration when the communication mechanism is computationally heavy. This low overhead is consistent with the lightweight design of ParaComm, especially its explicit linear reconstruction and Communication Backbone

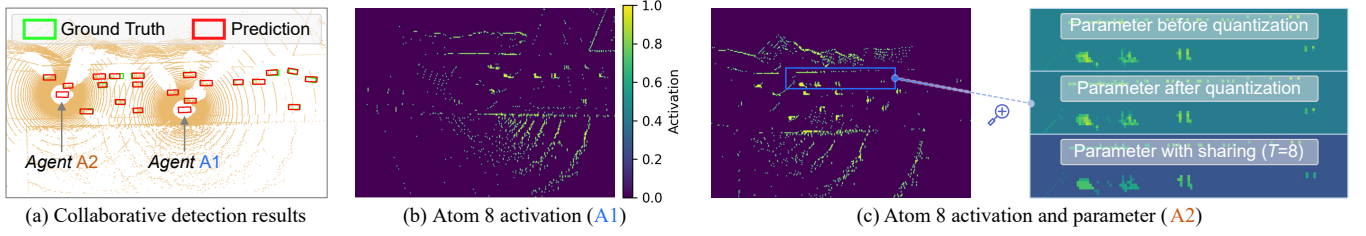


Fig. 12. Visualization of observation-dependent sparse parameter patterns. (a) Detection results in a two-agent scene. (b) The parameter activation pattern of agent A1. (c) The activation map and local parameter comparison of agent A2. The parameter patterns differ across agents, while the dominant activation layout remains stable after message compaction.

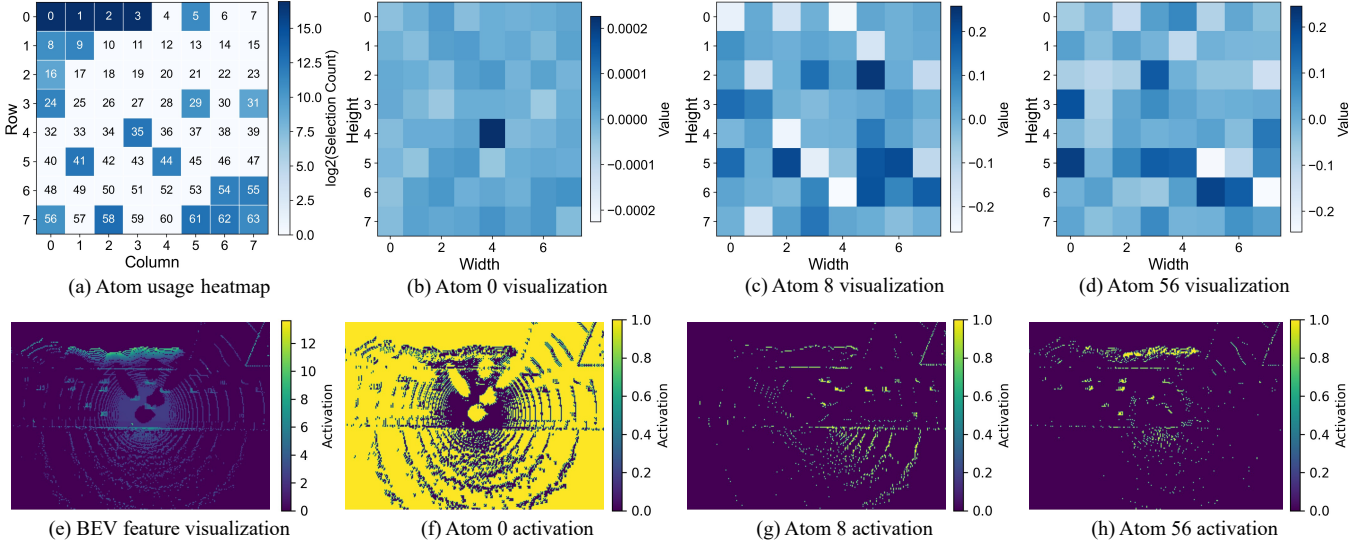


Fig. 13. Visualization of the interpretability of ParaComm. Atom usage is highly non-uniform, representative dictionary atoms exhibit distinct spatial patterns, and their activations respond to different regions of the BEV feature map. These results illustrate that ParaComm communicates sparse continuous parameters that selectively instantiate shared reusable bases under a given observation.

Sharing, which keep receiver-side processing simple.

7) *Analysis under Transmission Loss*: To evaluate ParaComm under transmission loss, we simulate transmission loss following the experimental protocol of Li et al. [50]. Specifically, each transmitted content item is randomly replaced with its mean with probability P , where P varies from 0.1 to 0.9 to represent progressively more severe communication degradation. Fig. 10 reports the AP@70 performance and absolute gain of Standard Collaboration and ParaComm under different transmission loss probabilities. Although both methods experience performance degradation as P increases, ParaComm consistently achieves higher AP@70 across the evaluated range, with an average improvement of 0.84% and a maximum gain of 1.67%. These results indicate that ParaComm is more tolerant of transmission loss and can better preserve detection performance when transmitted information is partially unavailable.

8) *Analysis under Non-Ideal Communication Conditions*: We further evaluate ParaComm under three non-ideal collaboration conditions, namely positional error, heading error, and time delay, following the V2X-ViT protocol [47]. Noise Level 0 denotes the clean setting, while Levels 1–5 correspond to increasingly severe perturbations. Fig. 11 reports the average AP gain of ParaComm over Standard Collaboration. Across all

perturbation types and levels, ParaComm consistently yields positive gains, ranging from 1.3 to 2.5 AP points. This indicates that the benefit of ParaComm is not confined to ideal settings, but persists under both localization uncertainty and temporal asynchrony. This behavior is consistent with ParaComm’s use of multiple shared atoms, where collaborative features are reconstructed from sparse activations over several atoms rather than a single transmitted representation. Such distributed composition may improve robustness to perturbations during collaborative perception.

9) *Generalization of the Shared Dictionary*: Following prior work [12], we examine whether the learned shared dictionary generalizes beyond the original training distribution. We train on the standard OPV2V set and evaluate on the Culver City Digital Town subset, which contains more realistic scene configurations. Tab. V shows that ParaComm remains the strongest method under this distribution shift and still outperforms Standard Collaboration. Although InfoCom is highly competitive in-distribution, it degrades more noticeably after the shift, and ParaComm surpasses it by an average of 1.64 percentage points. One possible explanation is that aggressive task-oriented filtering is more dependent on importance patterns learned from the source distribution, and may therefore remove information that becomes relevant under

shifted layouts and traffic patterns. In contrast, ParaComm keeps reusable bases in the shared dictionary and transmits only sparse observation-conditioned parameters, which may be less sensitive to such shifts.

10) *Analysis of the Advantages over Full-Feature Collaboration*: Standard Collaboration provides a strong full-feature reference, but transmitting all intermediate features is not necessarily optimal. Full feature maps may also carry redundant background responses, view-dependent artifacts, and noisy or misaligned activations that complicate cross-agent fusion. ParaComm’s effectiveness in alleviating these issues may stem from the regularization induced by its dictionary-based linear reconstruction and selective sparsification before transmission. Some empirical results support this interpretation. Tab.I shows gains across diverse collaboration settings, from DAIR-V2X and V2XSet to OPV2V with 2 to 7 agents, and the AP@70 improvements indicate that localization-sensitive accuracy is preserved. Tab.II and Fig. 11 further show consistent gains across base models and under position error, heading error, and time delay. These results suggest that ParaComm benefits from communicating compact parameters that lead to more selective and semantically regularized representations for fusion.

11) *Analysis of Sparse Parameter Patterns*: The parameter patterns in Fig. 12 vary across agents, indicating that the transmitted sparse parameters are conditioned on the current observation rather than collapsing to a shared generic template. Fig. 12a shows a scene observed by two agents, while Fig. 12b and Fig. 12c present the corresponding activation pattern and a local comparison before and after message compaction. In both agents, strong activations remain concentrated in regions emphasized by their respective viewpoints. Since the underlying dictionary atoms are shared and fixed, these cross-agent differences are expressed through the transmitted parameters rather than the semantic bases themselves. The local comparison further shows that quantization and Variable-Tile Parameter Sharing mainly affect numerical values, while the dominant activation layout remains visually stable. Overall, message compaction reduces value redundancy without erasing the spatial-atom structures and patterns.

12) *Interpretability Analysis of ParaComm*: An appealing property of ParaComm is that its transmitted message is easier to interpret than conventional feature-based communication. Our analysis is conducted from three perspectives: the frequency of atom usage, the structure of individual atoms, and their sample-specific activations. As shown in Fig. 13a, atom usage is clearly non-uniform, with some atoms being activated much more frequently than others. This reveals that the learned dictionary contains a set of reusable patterns that contribute differently across scenes. Figs. 13b–d further show that different atoms exhibit distinct spatial structures, indicating that they capture different components in the latent communication space. When the atoms are applied to a specific sample, the corresponding activation maps in Figs. 13f–h respond to different regions and scene structures relative to the BEV feature in Fig. 13e. Specifically, the activation in Fig. 13f is more strongly associated with background regions, while the atoms in Fig. 13g and Fig. 13h show stronger responses to objects located in front of and behind the vehicle, respectively.

These sparse activations indicate that only a limited number of atom-location pairs are needed to encode the collaborative information relevant to the current observation. Overall, the visualizations suggest ParaComm provides a more structured and interpretable communication form, where shared atoms and sparse parameters play complementary roles in representing reusable bases and observation-specific variations.

VI. CONCLUSION AND FUTURE WORK

In this paper, we revisited what to communicate in collaborative perception and argued that communication-efficient collaboration should not be limited to transmitting full feature representations. Instead, we introduced the perspective of Parameter-Centric Communication, which keeps reusable shared semantic bases local to each agent and communicates only the observation-specific variations needed for collaboration. Based on this idea, we proposed ParaComm, a learnable dictionary-driven framework that represents collaborative information with shared dictionary atoms and sparse continuous parameters. ParaComm further enables compact and effective communication through Joint Space-Atom Sparsification, Variable-Tile Parameter Sharing, Quantized Representation, Dictionary-Based Explicit Reconstruction, and Communication Backbone Sharing. Extensive experiments on multiple collaborative perception datasets demonstrate the effectiveness of ParaComm and highlight the practical value of Parameter-Centric Communication.

An important direction for future work is to move from fixed communication configurations to explicit online adaptation. Although ParaComm provides several communication-control variables, such as sparsity, tile size, and quantization bit-width, these variables remain fixed in the current implementation. Adapting the transmitted parameter budget to time-varying communication conditions and message availability could make ParaComm more flexible under dynamic collaboration. Beyond communication adaptation, another direction is to extend the shared dictionary to heterogeneous multi-class collaborative perception. The current evaluation mainly focuses on vehicle-oriented detection, whereas practical systems may involve pedestrians, trucks, and other traffic participants with diverse semantic and geometric patterns. Such object diversity may require larger-capacity and category-aware dictionary atoms to provide discriminative bases across object types. These promising extensions hold the potential to further broaden the applicability of ParaComm in practical collaborative perception.

REFERENCES

- [1] G. Gao, Y. Dong, R. Wang, and X. Zhou, “Edgevision: Towards collaborative video analytics on distributed edges for performance maximization,” *IEEE Transactions on Multimedia*, vol. 26, pp. 9083–9094, 2024.
- [2] Y. Han, H. Zhang, H. Zhang, C. Ding, Y. Cao, and Y. Li, “Cods: Enhancing collaborative perception in heterogeneous scenarios via domain separation,” *IEEE Transactions on Mobile Computing*, vol. 25, no. 3, pp. 4286–4299, 2026.
- [3] X. Gao, R. Xu, J. Li, Z. Wang, Z. Fan, and Z. Tu, “Stamp: Scalable task-and model-agnostic collaborative perception,” in *International Conference on Learning Representations (ICLR)*, 2025.

- [4] Q. Xia, W. Lin, H. Xiang, X. Huang, S. Chen, Z. Dong, C. Wang, and C. Wen, "Learning to detect objects from multi-agent lidar scans without manual labels," in *IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2025, pp. 1418–1428.
- [5] X. Huang, J. Wang, Q. Xia, S. Chen, B. Yang, X. Li, C. Wang, and C. Wen, "V2x-r: Cooperative lidar-4d radar fusion with denoising diffusion for 3d object detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 27390–27400.
- [6] S. Hu, Y. Tao, G. Xu, X. Qian, Y. Deng, X. Chen, S. T. W. Kwong, and Y. Fang, "Cp-uniguard: A unified, probability-agnostic, and adaptive framework for malicious agent detection and defense in multi-agent embodied perception systems," *IEEE Transactions on Mobile Computing*, pp. 1–14, 2026.
- [7] Y. Tao, S. Hu, H. An, Z. Fang, H. Cao, and Y. Fang, "Learning mutual view information graph for adaptive adversarial collaborative perception," in *IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2026.
- [8] Q. Wei, P. Dai, W. Li, B. Liu, and X. Wu, "Copeft: Fast adaptation framework for multi-agent collaborative perception with parameter-efficient fine-tuning," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 39, no. 22, 2025, pp. 23351–23359.
- [9] K. Yang, Z. Xu, D. Yang, Q. Fu, R. Tang, L. Song, and L. Zhang, "Robust multi-agent collaborative perception via spatio-temporal awareness," *IEEE Transactions on Circuits and Systems for Video Technology*, 2025.
- [10] J. Zha, Y. Fan, T. Zhang, G. Chen, Y. Chen, C. Gao, and X. Chen, "Aircopbench: A benchmark for multi-drone collaborative embodied perception and reasoning," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 2, 2026, pp. 1507–1515.
- [11] Q. Xie, X. Zhou, T. Hong, W. Hu, W. Qu, and T. Qiu, "Towards communication-efficient cooperative perception via planning-oriented feature sharing," *IEEE Transactions on Mobile Computing*, vol. 24, no. 4, pp. 2551–2563, 2025.
- [12] R. Xu, H. Xiang, X. Xia, X. Han, J. Liu, and J. Ma, "Opv2v: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication," in *International Conference on Robotics and Automation (ICRA)*, 2021, pp. 2583–2589.
- [13] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, "Where2comm: Communication-efficient collaborative perception via spatial confidence maps," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 4874–4886, 2022.
- [14] Q. Wei, P. Dai, W. Li, B. Liu, and X. Wu, "Infocom: Kilobyte-scale communication-efficient collaborative perception with information bottleneck," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 35, 2026, pp. 29731–29739.
- [15] Y. Hu, J. Peng, S. Liu, J. Ge, S. S. Liu, and S. Chen, "Communication-efficient collaborative perception via information filling with codebook," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 15481–15490.
- [16] Y. Hu, X. Pang, X. Qin, Y. C. Eldar, S. Chen, P. Zhang, and W. Zhang, "Pragmatic communication in multi-agent collaborative perception," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.
- [17] Z. Zhao, "Balf: Simple and efficient blur aware local feature detector," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 3362–3372.
- [18] S. Zeng, X. Chang, M. Xie, X. Liu, Y. Bai, Z. Pan, M. Xu, and X. Wei, "Futuresightdrive: Thinking visually with spatio-temporal cot for autonomous driving," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 38, pp. 67299–67318, 2026.
- [19] S. Zeng, X. Chang, X. Liu, Y. Yuan, S. Liang, Z. Pan, M. Xu, and X. Wei, "Priordrive: Enhancing online hd mapping with unified vector priors," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 15, 2026, pp. 12313–12321.
- [20] Z. Chen, K. You, J. Yang, L. Chen, F. Li, Z. Feng, and L. Jia, "A sparse-to-dense guided fusion framework for three-dimensional object detection in railway environments," *Engineering Applications of Artificial Intelligence*, vol. 178, p. 115095, 2026.
- [21] G. Liu, A. Hu, Y. Hu, W. Zhang, and S. Chen, "Rate-distortion optimized communication for collaborative perception," in *International Conference on Learning Representations (ICLR)*, 2026.
- [22] E. Bao and J. Yang, "Wavecomm: Lightweight communication for collaborative perception via wavelet feature distillation," in *International Conference on Robotics and Automation (ICRA)*, 2026.
- [23] J. Zha, C. Zhao, N. Zhou, Z. Liu, T. Sun, B. Zhang, X. Zhang, and X. Chen, "Breaking the communication-accuracy trade-off: A sparsified information diffusion framework for multi-agent collaborative perception," *IEEE Transactions on Mobile Computing*, 2026.
- [24] Q. Wei, P. Dai, K. Li, W. Li, X. Wu, and Z. Yu, "Cotrans: A unified framework for enhancing sparse-agent collaborative perception," *IEEE Transactions on Multimedia*, 2026.
- [25] K. Gong, P. Yao, G. Luo, Q. Yuan, T. Fu, H. Zhang, and J. Li, "From discriminative to generative: A diffusion-based paradigm for multi-agent collaborative perception," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 6, 2026, pp. 4266–4274.
- [26] Z. Fang, J. Wang, Y. Ma, Y. Tao, Y. Deng, X. Chen, and Y. Fang, "R-acp: Real-time adaptive collaborative perception leveraging robust task-oriented communications," *IEEE Journal on Selected Areas in Communications*, 2025.
- [27] H. Jiang, J. Guo, Z. Xiao, K. Yang, T. Li, and B. Li, "Collaborative perception and computing offloading in 6g air-ground integrated networks," *IEEE Journal on Selected Areas in Communications*, 2025.
- [28] Y. Li, Q. Yuan, X. Fu, G. Luo, Y. Yang, J. Kang, J. Li, and D. Niyato, "An efficient cross-agent spatial-temporal collaboration framework for environmental perception in iov," *IEEE Transactions on Cognitive Communications and Networking*, 2026.
- [29] G. Luo, C. Shao, N. Cheng, H. Zhou, H. Zhang, Q. Yuan, and J. Li, "Edgecooper: Network-aware cooperative lidar perception for enhanced vehicular awareness," *IEEE Journal on Selected Areas in Communications*, vol. 42, pp. 207–222, 2023.
- [30] J. Xu, Y. Zhang, Z. Cai, and D. Huang, "Cosdh: Communication-efficient collaborative perception via supply-demand awareness and intermediate-late hybridization," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6834–6843, 2025.
- [31] Y. Tao, S. Hu, Y. Hu, H. An, H. Cao, and Y. Fang, "Gcp: guarded collaborative perception with spatial-temporal aware malicious agent detection," *IEEE Transactions on Dependable and Secure Computing*, 2026.
- [32] K. Yang, D. Yang, J. Zhang, H. Wang, P. Sun, and L. Song, "What2comm: Towards communication-efficient collaborative perception via feature decoupling," in *ACM International Conference on Multimedia (ACM MM)*, 2023, pp. 7686–7695.
- [33] R. Wang, X. Gao, H. Xiang, R. Xu, and Z. Tu, "Cocmt: Communication-efficient cross-modal transformer for collaborative perception," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [34] J. Wang, Z. Jiang, W. Sun, J. Zhong, H. Yu, Y. Zhang, C. Lu, C. Zhang, L. He, S. Xu *et al.*, "Sparsecoop: Cooperative perception with kinematic-grounded queries," in *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 12, 2026, pp. 9876–9884.
- [35] Y.-C. Liu, J. Tian, N. Glaser, and Z. Kira, "When2com: Multi-agent perception via communication graph grouping," in *IEEE/CVF Conference on computer vision and pattern recognition (CVPR)*, 2020, pp. 4106–4115.
- [36] Y.-C. Liu, J. Tian, C.-Y. Ma, N. Glaser, C.-W. Kuo, and Z. Kira, "Who2com: Collaborative perception via learnable handshake communication," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 6876–6883.
- [37] J. Zhang, K. Yang, Y. Wang, H. Wang, P. Sun, and L. Song, "Er-mvp: Communication-efficient and collaboration-robust multi-vehicle perception in challenging environments," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 12575–12584.
- [38] D. Yang, K. Yang, Y. Wang, J. Liu, Z. Xu, R. Yin, P. Zhai, and L. Zhang, "How2comm: Communication-efficient and collaboration-pragmatic multi-agent perception," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, pp. 25151–25164, 2023.
- [39] J. Zhou, P. Dai, Q. Wei, B. Liu, X. Wu, and J. Wang, "Pragmatic heterogeneous collaborative perception via generative communication mechanism," *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [40] C. Liu, Z. Chao, and S. Chen, "Linking modality isolation in heterogeneous collaborative perception," in *IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2026.
- [41] M. Aharon, M. Elad, A. M. Bruckstein, and Y. Katz, "K-svd: An algorithm for designing of overcomplete dictionaries for sparse representation," *IEEE Transactions on Signal Processing*, vol. 54, no. 11, p. 4311–4322, 2006.
- [42] P. G. Daneshmand and H. Rabbani, "Tensor ring decomposition guided dictionary learning for oct image denoising," *IEEE Transactions on Medical Imaging*, vol. 43, pp. 2547–2562, 2024.
- [43] L. Zhang, Y. Li, X. Zhou, X. Zhao, and S. Gu, "Transcending the limit of local window: Advanced super-resolution transformer with adaptive token dictionary," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2856–2865, 2024.

- [44] X. Zhang, N. Li, and A. Dai, “Dnf: Unconditional 4d generation with dictionary-based neural fields,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 26 047–26 056.
- [45] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A convnet for the 2020s,” in *IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2022, pp. 11 976–11 986.
- [46] Y. Lu, Q. Li, B. Liu, M. Dianat, C. Feng, S. Chen, and Y. Wang, “Robust collaborative 3d object detection in presence of pose errors,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 4812–4818.
- [47] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, “V2x-vit: Vehicle-to-everything cooperative perception with vision transformer,” in *European Conference on Computer Vision (ECCV)*, 2022, pp. 107–124.
- [48] H. Yu, Y. Luo, M. Shu, Y. Huo, Z. Yang, Y. Shi, Z. Guo, H. Li, X. Hu, J. Yuan *et al.*, “Dair-v2x: A large-scale dataset for vehicle-infrastructure cooperative 3d object detection,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 21 361–21 370.
- [49] Z. Li, H. Liang, H. Wang, M. Zhao, J. Wang, and X. Zheng, “Mkd-cooper: Cooperative 3d object detection for autonomous driving via multi-teacher knowledge distillation,” *IEEE Transactions on Intelligent Vehicles*, pp. 1490–1500, 2024.
- [50] J. Li, R. Xu, X. Liu, J. Ma, Z. Chi, J. Ma, and H. Yu, “Learning for vehicle-to-vehicle cooperative perception under lossy communication,” *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 4, pp. 2650–2660, 2023.



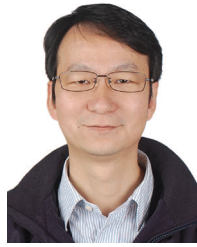
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